A Stackelberg Security Game Formulation for Aircraft Patrol Planning

Theory, Linear Programming Formulation, and Implementation

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Contents

1 Introduction **3**

2 Problem Parameters **3**
2.1 Fleet and Barriers . . . 3

3 Coverage Model **4**

4 The Stackelberg Security Game **4**
4.1 Players and Actions . . . 4
4.2 Payoffs . . . 4
4.3 Stackelberg Equilibrium . . . 5

5 Stage 1: Linear Programme and Analytical Solution **5**
5.1 LP Formulation . . . 5
5.2 Analytical Solution . . . 5
5.3 Active and Inactive Barrier Types . . . 6

6 Stage 2: Mixed Strategy via Feasibility LP **7**
6.1 From Continuous to Integer Schedules . . . 7
6.2 Feasibility LP . . . 7
6.3 Solving the Feasibility LP via Column Generation . . . 7
6.4 Generating Candidate Strategies . . . 9

7 Stage 3: Scheduling **9**
7.1 Distributing Sorties Across Individual Barriers . . . 9
7.2 Randomised Phase Allocation . . . 9
7.3 Aircraft Assignment . . . 10

8 Minimum Fleet Size **10**

9 Worked Example: Toy Problem **11**
9.1 Parameters . . . 11
9.2 Stage 1: Analytical Solution . . . 11
9.3 Stage 2: Mixed Strategy . . . 12
9.4 Stage 3: Concrete Timetables . . . 12
9.5 Game Value Verification . . . 13
9.6 Minimum Fleet Size . . . 13

10 Solution Algorithm **14**

11 Why the Floor/Ceil Mixed Strategy is Worst-Case Optimal **14**
11.1 Definitions . . . 14
11.2 The Key Lemma . . . 15
11.3 Main Theorem . . . 15
11.4 Why Diversification Cannot Help . . . 16

12 Comparison with Baseline Patrol Policies **16**
12.1 Baseline 1: Fixed Deterministic Patrol . . . 16
12.2 Baseline 2: Uniform Random Patrol . . . 17
12.3 Baseline 3: Value-Proportional Patrol . . . 17
12.4 Baseline 4: Greedy (Highest-Value First) . . . 18
12.5 Summary Table and Discussion . . . 18

1 Introduction

We consider a patrol planning problem in which a defender allocates a fleet of aircraft to protect a set of barriers against an intelligent attacker. The attacker observes the defender's patrol *policy* (but not today's specific schedule) and selects the most vulnerable target. This interaction is modelled as a *Stackelberg security game* [1], in which the defender commits to a mixed strategy as the *leader* and the attacker responds optimally as the *follower*.

The solution pipeline has three stages:

Stage 1. Solve a linear programme (LP) to obtain the optimal flying-hour allocation h^* per barrier type.

Stage 2. Solve a feasibility LP to express h^* as a convex combination of integer sortie schedules, yielding a mixed strategy s^*.

Stage 3. Translate each pure strategy in the support of s^* into a concrete daily timetable with explicit aircraft assignments.

Additionally, we characterise the *minimum fleet size* required for zero expected loss and for guaranteed full coverage.

2 Problem Parameters

2.1 Fleet and Barriers

Table 1: Problem parameters

Symbol	Meaning	Units
N	Number of aircraft in the fleet	aircraft
K	Number of barrier types	—
n_k	Number of individual barriers of type k	barriers
v_k	Military value of one barrier of type k	value units
N_b	Total number of barriers $= \sum_{k=1}^{K} n_k$	barriers
T	Planning horizon (schedule repeats periodically)	hours
τ_p	Patrol duration (flying hours per sortie)	hours
τ_s	Safe period after patrol ends	hours
W	Protection window $= \tau_p + \tau_s$	hours
μ	Window multiplier $= W/\tau_p$	—
τ_r	Aircraft restart period (turnaround after landing)	hours
G	Minimum gap between consecutive sortie starts for the same aircraft $= \tau_p + \tau_r$	hours
F	Maximum flying hours per aircraft per period T	hours
B	Total fleet flying hours $= N \cdot F$	hours
N_{srt}	Maximum integer sorties $= \lfloor B/\tau_p \rfloor$	sorties
B_{eff}	Integer-feasible budget $= N_{\text{srt}} \cdot \tau_p$	hours

Remark 1. *$B_{\text{eff}} \leq B$ with a gap of at most $\tau_p - 1$ hours (less than one sortie). Stage 1 is solved with B_{eff} so that Stage 2 closes with zero residual gap.*

3 Coverage Model

Let h_k denote the total flying hours allocated per period T to *all* barriers of type k collectively. Each sortie of τ_p flying hours protects a barrier for a full window of $W = \mu\,\tau_p$ hours. Hence, h_k flying hours generate $\mu\,h_k$ protection hours distributed across the n_k barriers of type k.

Definition 1 (Coverage probability)**.** *The coverage probability of a specific barrier of type k – the fraction of the planning horizon during which the barrier is protected – is:*

$$P_k \;=\; \frac{\mu\,h_k}{n_k\,T}. \tag{1}$$

This model assumes sorties are distributed uniformly in time across all barriers of the same type, which is enforced in Stage 3.

Remark 2. *$P_k \in [0,1]$ requires $h_k \le n_k\,T/\mu$. The case $P_k = 1$ (full coverage) corresponds to every barrier of type k being continuously protected, which requires $h_k = n_k\,T/\mu$ flying hours.*

4 The Stackelberg Security Game

4.1 Players and Actions

The game has two players: the **defender** (fleet operator) and the **attacker**. The defender moves first by committing to a mixed patrol policy. The attacker observes this policy and selects a target barrier to strike.

Definition 2 (Pure strategy)**.** *A* pure patrol strategy *is an integer sortie allocation $(m_1, \ldots, m_K)$ with $m_k \ge 0$ sorties on barriers of type k and $\sum_k m_k = N_{srt}$. The corresponding flying-hour allocation is $a_k = m_k\,\tau_p$.*

Definition 3 (Mixed strategy)**.** *A* mixed patrol strategy *is a probability distribution $s = (s_1, \ldots, s_J)$ over a finite set of J pure strategies, with $s_j \ge 0$ and $\sum_j s_j = 1$. The expected flying-hour allocation under s is $h_k = \sum_j A_{jk}\,s_j$, where A_{jk} is the flying hours on type k in pure strategy j.*

4.2 Payoffs

If the attacker strikes a barrier of type k, the zero-sum payoff to the defender is:

$$u_d(k) \;=\; v_k \cdot P_k - v_k \cdot (1 - P_k) \;=\; v_k\,(2P_k - 1),$$

reflecting a gain of v_k if the barrier is covered (probability P_k) and a loss of v_k if not (probability $1 - P_k$). The attacker's payoff is $u_a(k) = -u_d(k)$.

Substituting (1):

$$U_d(k;\,h) \;=\; v_k\left(\frac{2\mu\,h_k}{n_k\,T} - 1\right). \tag{2}$$

4.3 Stackelberg Equilibrium

The defender's problem is:

$$\max_{h \geq 0,\, \sum_k h_k = B_{\text{eff}}} \; \min_{k \in \{1,\ldots,K\}} U_d(k;\, h). \tag{3}$$

The inner minimisation represents the attacker choosing the worst barrier for the defender. The outer maximisation is the defender's commitment.

Remark 3 (Why the attacker must be indifferent at optimum)**.** *Suppose at optimum $U_d(k') > U_d(k'')$ for some pair k', k''. The attacker always strikes k'' (lower defender utility), giving a game value of $U_d(k'')$. But the defender can then shift hours from k' to k'', raising $U_d(k'')$ without decreasing the worst case, strictly improving the objective. This contradicts optimality. Hence at the optimum, all types in the* active support $\mathcal{A} = \{k : h_k^* > 0\}$ *must satisfy $U_d(k;\, h^*) = z^*$ for a common value z^* (the* game value*). Types outside $\mathcal{A}$ satisfy $U_d(k;\, h^*) \geq z^*$ as a post-optimality consistency check.*

5 Stage 1: Linear Programme and Analytical Solution

5.1 LP Formulation

Problem (3) is reformulated as a linear programme by introducing a scalar variable z (the game value):

$$\left|\begin{array}{lll} \max\limits_{h_1,\ldots,h_K,\,z} & z & \\ \text{s.t.} & z \leq v_k\left(\dfrac{2\mu\, h_k}{n_k\, T} - 1\right), \quad k = 1,\ldots,K & \text{(C1)} \\ & \displaystyle\sum_{k=1}^{K} h_k = B_{\text{eff}} & \text{(C2)} \\ & h_k \geq 0, \quad k = 1,\ldots,K & \text{(C3)} \end{array}\right. \tag{4}$$

The LP has $K{+}1$ variables $(h_1, \ldots, h_K, z)$ and $K{+}2$ constraints (before non-negativity). It is always feasible and bounded, so a finite optimum exists.

5.2 Analytical Solution

Theorem 1 (Optimal coverage hours)**.** *Assuming all K types are active at optimum (verified post-hoc via the consistency check), the optimal solution to LP (4) is:*

$$z^* \;=\; \frac{B_{\mathit{eff}} - \dfrac{TN_b}{2\mu}}{\displaystyle\sum_{k=1}^{K} d_k}, \tag{5}$$

$$h_k^* \;=\; d_k\,(z^* + v_k), \quad k = 1,\ldots,K, \tag{6}$$

where

$$d_k \;=\; \frac{n_k\, T}{2\,\mu\, v_k} \tag{7}$$

and $N_b = \sum_k n_k$ *is the total number of barriers.*

Proof. At optimum all K constraints (C1) bind, giving $U_d(k;\, h^*) = z^*$ for each k:

$$v_k\left(\frac{2\mu\, h_k^*}{n_k T} - 1\right) = z^* \;\;\Longrightarrow\;\; h_k^* \;=\; \frac{n_k T}{2\mu v_k}(z^* + v_k) \;=\; d_k(z^* + v_k).$$

Substituting into (C2):

$$\sum_{k=1}^{K} d_k(z^* + v_k) = B_{\text{eff}} \;\;\Longrightarrow\;\; z^* \sum_k d_k + \sum_k d_k v_k = B_{\text{eff}}.$$

Since $d_k v_k = n_k T/(2\mu)$, we have $\sum_k d_k v_k = T N_b/(2\mu)$. Solving for z^* gives (5). □ □

Remark 4 (Sign of z^*)**.** $z^* > 0$ *iff* $B_{\text{eff}} > T N_b/(2\mu)$*, i.e. the fleet's total protection capacity exceeds half the total barrier-hours. When* $z^* > 0$ *the defender has a net expected gain; when* $z^* < 0$ *every attack yields a net expected loss for the defender despite optimal patrol.*

5.3 Active and Inactive Barrier Types

Since $d_k > 0$ and $v_k > 0$, the sign of $h_k^* = d_k(z^* + v_k)$ is determined entirely by whether $z^* + v_k$ is positive, zero, or negative. Three cases arise:

Case 1. $z^* > 0$ **(defender net gain).** Then $z^* + v_k > v_k > 0$ for every type k, so $h_k^* > 0$ for all k. The defender allocates positive patrol time to every barrier type without exception. This is the regime in which the fleet is large enough that the defender has a net advantage.

Case 2. $-v_k < z^* \le 0$ **(defender loss, but type** k **still worth defending).** The formula still gives $h_k^* = d_k(z^* + v_k) > 0$. Even though the defender suffers a net expected loss, abandoning type k would hand the attacker a free gain of $v_k > |z^*|$, which is strictly worse than the current loss. Type k remains active.

Case 3. $z^* \le -v_k$ **(type** k **indefensible).** The formula gives $h_k^* \le 0$, which violates non-negativity. Type k must be excluded from the active set. The defender's budget is so limited that even an undefended type k (with attacker gain v_k) is no worse for the defender than the equilibrium loss $|z^*|$ already sustained on the remaining types.

Proposition 2 (Active type characterisation)**.** *The active set at the Stackelberg equilibrium is* $\mathcal{A} = \{k : z^* > -v_k\}$*. All types are active whenever* $z^* \ge 0$*. A type* k *is inactive only when* $z^* < 0$ *(a loss situation)* and $|z^*| \ge v_k$ *(the equilibrium loss already equals or exceeds the full value of that type).*

Remark 5. *In practice, if the analytical formula gives any* $h_k^* < 0$*, we remove that type from* $\mathcal{A}$*, reduce* B_{eff} *by the hours that would have gone to it (zero, since it is excluded), and re-solve (5) over the reduced active set. The LP automatically returns* $h_k^* = 0$ *for inactive types, but the analytical formula requires this manual identification step.*

6 Stage 2: Mixed Strategy via Feasibility LP

6.1 From Continuous to Integer Schedules

The optimal flying-hour allocation h^* is a continuous vector. In practice, sorties are discrete: the number of sorties on type k must be a non-negative integer m_k. Define the *integer sortie target* as $m_k^* = h_k^*/\tau_p$. In general m_k^* is non-integer, so no single pure strategy achieves h^* exactly. We must mix two or more integer strategies.

6.2 Feasibility LP

Let $\mathcal{J}$ be the set of candidate pure strategies (integer sortie allocations). The coverage matrix $A \in \mathbb{R}^{|\mathcal{J}| \times K}$ has entries $A_{jk} = m_{jk}\,\tau_p$ (flying hours on type k in strategy j). We seek mixing probabilities $s \in \mathbb{R}^{|\mathcal{J}|}$ satisfying:

$$A^\top s \;=\; h^*, \quad \mathbf{1}^\top s = 1, \quad s \geq 0. \tag{8}$$

This is a *pure feasibility* problem with no objective. We convert it to an optimisation problem using a Phase-1 slack vector $\varepsilon \in \mathbb{R}^K_{\geq 0}$ measuring coverage deficit per barrier type:

$$\left|\begin{array}{lll} \min\limits_{s,\,\varepsilon} & \mathbf{1}^\top \varepsilon & \\ \text{s.t.} & A^\top s + \varepsilon = h^* & \text{(F1)} \\ & \mathbf{1}^\top s = 1 & \text{(F2)} \\ & s \geq 0, \quad \varepsilon \geq 0 & \text{(F3)} \end{array}\right. \tag{9}$$

A zero optimal value ($\mathbf{1}^\top \varepsilon^* = 0$) confirms that h^* is achievable as a convex combination of the candidate strategies.

6.3 Solving the Feasibility LP via Column Generation

Problem (9) is solved using the standard column generation algorithm. The total number of feasible integer strategies—all non-negative integer vectors $(m_1, \ldots, m_K)$ summing to N_{srt}—is combinatorially large, so we never form the full matrix A explicitly. Instead, we maintain a small *restricted* column set $\mathcal{J}$, solve (9) over it, and add one new column per iteration until the gap $\mathbf{1}^\top \varepsilon^*$ reaches zero.

LP duality background. Every LP $\min c^\top x$ s.t. $Ax = b$, $x \geq 0$ has a *dual LP* $\max b^\top y$ s.t. $A^\top y \leq c$, where $y \in \mathbb{R}^m$ is a free variable vector associated one-to-one with the m equality constraints of the primal. At any feasible primal–dual pair (x, y), the *reduced cost* of primal variable x_j is

$$\bar{c}_j \;=\; c_j - a_j^\top y, \tag{10}$$

where a_j is column j of A. The **complementary slackness** conditions state that at optimality:

$$x_j > 0 \implies \bar{c}_j = 0, \qquad \bar{c}_j > 0 \implies x_j = 0.$$

Equivalently, a variable x_j not yet in the LP can improve the objective if and only if $\bar{c}_j < 0$. If no such variable exists, the current solution is already optimal.

Dual variables for (9). Assign dual variable $\lambda_k \in \mathbb{R}$ to each coverage constraint (F1) and $\nu \in \mathbb{R}$ to the probability constraint (F2). The costs are $c_{s_j} = 0$ for each strategy s_j and $c_{\varepsilon_k} = 1$ for each slack ε_k. Dual feasibility therefore requires:

$$a_j^\top \lambda + \nu \;\leq\; 0 \quad \forall j \in \mathcal{J}, \qquad \lambda_k \;\leq\; 1 \quad \forall k.$$

The reduced cost of adding a new strategy $j \notin \mathcal{J}$ is:

$$\bar{c}_j \;=\; 0 - (a_j^\top \lambda + \nu) \;=\; -(a_j^\top \lambda + \nu).$$

A new strategy improves the objective (reduces the total coverage gap) if and only if $\bar{c}_j < 0$, i.e. $a_j^\top \lambda + \nu > 0$.

Interpretation of λ_k. At any intermediate solution, $\lambda_k \geq 0$ measures the marginal reduction in total slack $\mathbf{1}^\top \varepsilon$ per additional hour of coverage on type k. A large λ_k means type k is the most under-covered and most urgently needs additional patrol hours. When the gap reaches zero, all $\varepsilon_k = 0$ and complementary slackness forces $\lambda_k \leq 1$ with no residual signal.

The pricing problem. At each iteration, we solve:

$$j^* \;=\; \underset{m_k \geq 0,\, \sum_k m_k = N_{\text{srt}}}{\arg\max} \quad \tau_p \sum_{k=1}^{K} \lambda_k \, m_k. \tag{11}$$

This asks: given the current under-coverage signal λ, which integer sortie allocation fills the gap most efficiently? Crucially, (11) does *not* require knowing all feasible strategies. Because every sortie costs exactly τ_p flying hours and contributes $\tau_p \lambda_k$ to the objective when assigned to type k, the optimal solution is to assign all N_{srt} sorties to the type with the highest λ_k, distributing any remaining sorties to the next highest type if an integer split is needed. This is a simple sort, with a closed-form answer.

Column generation algorithm.

1. **Initialise** $\mathcal{J}$ with K canonical strategies (strategy k assigns all N_{srt} sorties to type k).
2. **Solve** the restricted feasibility LP (9) over $\mathcal{J}$. Obtain primal (s^*, ε^*) and dual (λ^*, ν^*).
3. **If** $\mathbf{1}^\top \varepsilon^* = 0$: **stop**. The current s^* is the desired mixed strategy.
4. **Solve** the pricing problem (11) using λ^* to find j^*.
5. **Add** column a_{j^*} to $\mathcal{J}$ and return to step 2.

By Proposition 3, the algorithm terminates in at most $K+1$ iterations, since the feasibility LP needs at most $K+1$ columns in its optimal basis. In practice, for the problem sizes considered here ($K \leq 4$), the gap reaches zero after two or three columns are added, and the full column generation loop is replaced by direct enumeration of the floor/ceil neighbourhood of $m^* = h^*/\tau_p$ as described in Section 6.4.

6.4 Generating Candidate Strategies

The minimal set of candidates is obtained by rounding each m_k^* to its floor and ceiling, then deriving the remaining type counts to satisfy the total sortie constraint $\sum_k m_k = N_{\text{srt}}$. This yields at most 2^{K-1} candidate strategies, which is small for $K \leq 4$. Additional perturbations (± 1 on each type, compensated on another) are added for robustness.

Proposition 3 (Support size bound)**.** *The Phase-1 LP (9) has $K+1$ equality constraints: K coverage constraints (F1) and one probability constraint (F2). Any linear programme returns a basic feasible solution, which has at most as many nonzero variables as equality constraints. Therefore the LP solver always returns a mixing distribution s^* with at most $K+1$ pure strategies in its support, regardless of the total number of candidate strategies $|\mathcal{J}|$.*

Concretely: with $K = 2$ barrier types, at most 3 pure strategies suffice; with $K = 3$, at most 4. In the toy example of Section 9 ($K = 2$), exactly 2 strategies appear in the support because the system has a unique solution.

7 Stage 3: Scheduling

7.1 Distributing Sorties Across Individual Barriers

Given the sortie count m_k for type k in a pure strategy, distribute sorties across the n_k individual barriers as uniformly as possible. Let $q_i = \lfloor m_k/n_k \rfloor + \mathbf{1}[i < (m_k \bmod n_k)]$ be the sortie count for barrier i of type k. Each barrier then has a patrol interval of:

$$\Delta_i \;=\; \frac{T}{q_i}. \tag{12}$$

The resulting coverage probability for individual barrier i is:

$$p_i \;=\; \frac{W}{\Delta_i} \;=\; \frac{W\,q_i}{T} \;=\; \frac{\mu\,\tau_p\,q_i}{T}. \tag{13}$$

7.2 Randomised Phase Allocation

The patrol *frequency* for barrier i is fixed at one sortie every Δ_i hours by the sortie count q_i derived in Stage 2. However, fixing the absolute start times—so that barrier i is always patrolled at the same hours every day—creates a predictable gap that an observant adversary can exploit. If the patrol clock is known, the adversary simply waits until just after a patrol ends and attacks in the unprotected window before the next one.

To prevent this, each barrier draws its own independent phase offset at the start of each day. Specifically, barrier i draws:

$$\phi_i \;\sim\; \text{Uniform}\big[0,\; \Delta_i - W\big), \tag{14}$$

where $W = \tau_p + \tau_s$ is the protection window. The j-th patrol start time on that day is then:

$$t_{i,j} \;=\; \phi_i \;+\; j\,\Delta_i, \quad j = 0, 1, 2, \ldots \tag{15}$$

Remark 6 (Why $\Delta_i - W$ and not Δ_i)**.** *The upper limit $\Delta_i - W$ ensures the full protection window $[\phi_i,\, \phi_i + W]$ fits within the current Δ_i-length cycle without overlapping into the next*

cycle. This keeps the coverage model exact: exactly W out of every Δ_i hours are protected, with no double-counting across cycle boundaries. The condition $\Delta_i > W$ (required for the interval to be non-empty) is always satisfied here since $\Delta_i \geq 720/41 \approx 17.6\,h$ while $W = 9\,h$.

Remark 7 (Coverage is preserved)**.** *The coverage probability $p_i = W/\Delta_i$ depends only on the ratio of window to interval, not on the phase ϕ_i. In each cycle of length Δ_i, exactly W hours are protected regardless of where ϕ_i falls. Randomising ϕ_i therefore leaves the Stackelberg game value z^* and all coverage guarantees unchanged.*

Remark 8 (Why individual draws are more unpredictable than shared draws)**.** *An earlier formulation used a single shared phase draw ϕ_k per barrier* type*, with individual barriers staggered deterministically from it by multiples of Δ_i/n_k. Under that scheme, observing one barrier's departure time reveals the phases of all other barriers of the same type. Under the individual draw in (14), each barrier's phase is an independent uniform variable. An adversary must infer n_k independent phases rather than one, making timing prediction significantly harder, while the coverage guarantee is identical.*

Remark 9 (Two layers of randomisation)**.** *Under mixed-strategy randomisation (Stage 2), the adversary knows the probabilities $(s_1, \ldots, s_J)$ of the monthly plans but not which plan is active this month. Under phase randomisation (Stage 3), even if the adversary identifies the active monthly plan, they still cannot predict the start times of individual sorties, because ϕ_i is drawn fresh each day and never revealed. The two layers protect against different information threats: Stage 2 protects the* sortie allocation*; Stage 3 protects the* patrol timing*.*

7.3 Aircraft Assignment

Aircraft are grouped by barrier type. Within each group, sorties are assigned in chronological order using a greedy round-robin, subject to the gap constraint: consecutive sorties of the same aircraft must be separated by at least $G = \tau_p + \tau_r$ hours (patrol duration plus turnaround). The budget constraint F hours per aircraft per period is verified for each assigned aircraft.

8 Minimum Fleet Size

Definition 4 (Coverage threshold)**.** *For a desired coverage level $\theta \in [0, 1]$, we require $P_k \geq \theta$ for all $k = 1, \ldots, K$.*

From (1), $P_k \geq \theta$ requires $h_k \geq \theta\, n_k T/\mu$. Summing over all types:

$$B_{\text{eff}} \;\geq\; \frac{\theta\, T}{\mu} \sum_{k=1}^{K} n_k \;=\; \frac{\theta\, T\, N_b}{\mu}.$$

The minimum number of aircraft is therefore:

$$N_{\min}(\theta) \;=\; \left\lceil \frac{\theta\, T\, N_b}{\mu\, F} \right\rceil. \tag{16}$$

Two critical thresholds:

$$\theta = \tfrac{1}{2}: \quad N_{\min}(\tfrac{1}{2}) = \left\lceil \frac{TN_b}{2\mu F} \right\rceil \quad \text{(zero expected loss, } z^* \geq 0\text{)}, \tag{17}$$

$$\theta = 1: \quad N_{\min}(1) = \left\lceil \frac{TN_b}{\mu F} \right\rceil \quad \text{(guaranteed full coverage)}. \tag{18}$$

Remark 10 (Equivalence with capacity argument)**.** *Equation* (18) *is equivalent to the simple capacity argument: one aircraft produces F/τ_p sorties per period, each protecting for W hours, giving $F\mu/\tau_p \cdot \tau_p = \mu F$ protection-hours per aircraft. Full coverage of N_b barriers over T hours requires $N_b T$ protection-hours in total, hence $N_{min}(1) = \lceil N_b T/(\mu F) \rceil$.*

9 Worked Example: Toy Problem

We illustrate the method with a small, fully tractable instance in which every intermediate quantity is an exact rational number.

9.1 Parameters

Table 2: Toy problem parameters

Parameter	Value	Notes
N	2	aircraft
K	2	barrier types: H (High), M (Medium)
n_H, n_M	1, 1	one barrier of each type
v_H, v_M	6, 2	value units
T	24 h	planning horizon (repeats daily)
τ_p	2 h	patrol duration
τ_s	4 h	safe period
$W = \mu\,\tau_p$	6 h, $\mu = 3$	protection window
τ_r	4 h	restart period (turnaround)
$G = \tau_p + \tau_r$	6 h	minimum gap between sortie starts
F	6 h/aircraft/day	flying budget
$B = NF$	12 h	fleet total
$N_{\text{srt}} = B/\tau_p$	6	sorties per day (exact integer)
B_{eff}	12 h	(equal to B here)

9.2 Stage 1: Analytical Solution

Computing d_k from (7):

$$d_H = \frac{1 \times 24}{2 \times 3 \times 6} = \frac{24}{36} = \frac{2}{3}, \qquad d_M = \frac{1 \times 24}{2 \times 3 \times 2} = \frac{24}{12} = 2.$$

Applying Theorem 1:

$$z^* = \frac{12 - \frac{24 \times 2}{2 \times 3}}{\frac{2}{3} + 2} = \frac{12 - 8}{\frac{8}{3}} = \frac{4 \times 3}{8} = \frac{3}{2}.$$

$$h_H^* = \tfrac{2}{3}\big(\tfrac{3}{2}+6\big) = \tfrac{2}{3}\times\tfrac{15}{2} = 5 \text{ h}, \qquad h_M^* = 2\big(\tfrac{3}{2}+2\big) = 2\times\tfrac{7}{2} = 7 \text{ h}.$$

Budget check: $5+7=12=B_{\text{eff}}$. ✓

Coverage probabilities:

$$P_H = \frac{3\times 5}{1\times 24} = \frac{5}{8} = 0.625, \qquad P_M = \frac{3\times 7}{1\times 24} = \frac{7}{8} = 0.875.$$

Payoff verification:

$$U_d(H) = 6\big(2\times\tfrac{5}{8}-1\big) = 6\times\tfrac{1}{4} = \tfrac{3}{2} = z^*, \qquad U_d(M) = 2\big(2\times\tfrac{7}{8}-1\big) = 2\times\tfrac{3}{4} = \tfrac{3}{2} = z^*.$$

Both constraints (C1) bind: the attacker is indifferent. ✓

9.3 Stage 2: Mixed Strategy

Integer sortie targets: $m_H^* = 5/2 = 2.5$, $m_M^* = 7/2 = 3.5$ (both non-integer). The two bounding strategies are:

$$\text{Strategy A: } (m_H, m_M) = (3,3) \;\Rightarrow\; a_H = 6,\ a_M = 6.$$

$$\text{Strategy B: } (m_H, m_M) = (2,4) \;\Rightarrow\; a_H = 4,\ a_M = 8.$$

The feasibility system (8) reduces to:

$$6\,s_A + 4\,s_B = 5, \quad 6\,s_A + 8\,s_B = 7, \quad s_A + s_B = 1.$$

Subtracting the first from the second: $4\,s_B = 2 \Rightarrow s_B = \frac{1}{2}$, $s_A = \frac{1}{2}$.

Operational prescription: Flip a fair coin at the start of each day. Heads → Strategy A; Tails → Strategy B.

9.4 Stage 3: Concrete Timetables

Strategy A: (3H, 3M)

H barrier: 3 sorties spaced $24/3 = 8$ h apart, starting at 00:00. M barrier: 3 sorties spaced $24/3 = 8$ h apart, offset by 4 h $= \tau_p$, starting at 04:00.

Table 3: Strategy A timetable (AC-1 flies all H sorties, AC-2 flies all M)

Aircraft	Type	Depart	Safe until
AC-1	H	00:00	06:00
AC-2	M	04:00	10:00
AC-1	H	08:00	14:00
AC-2	M	12:00	18:00
AC-1	H	16:00	22:00
AC-2	M	20:00	02:00

Each aircraft flies $3\times 2 = 6$ h $= F$. ✓Minimum gap per aircraft: 8 h $\geq G = 6$ h. ✓

Coverage timelines (X = protected, · = exposed):

```
H: XXXXXX..XXXXXX..XXXXXX..    (18/24 h, P_H = 0.75)
M: XX..XXXXXX..XXXXXX..XXXX    (18/24 h, P_M = 0.75, wraps at midnight)
Hr: 0    4    8    12   16   20   24
```

Strategy B: (2H, 4M)

H barrier: 2 sorties at 00:00 and 12:00 (12 h apart). M barrier: 4 sorties at 03:00, 09:00, 15:00, 21:00 (6 h apart).

Table 4: Strategy B timetable

Aircraft	Type	Depart	Safe until
AC-1	H	00:00	06:00
AC-2	M	03:00	09:00
AC-1	M	09:00	15:00
AC-2	H	12:00	18:00
AC-1	M	15:00	21:00
AC-2	M	21:00	03:00

Each aircraft flies $3 \times 2 = 6$ h $= F$. ✓Minimum gap (AC-1): 09:00 − 00:00 = 9 h $\geq G$; 15:00 − 09:00 = 6 h $= G$. ✓Minimum gap (AC-2): 12:00 − 03:00 = 9 h; 21:00 − 12:00 = 9 h. ✓

Coverage timelines:

```
H: XXXXXX......XXXXXX......   (12/24 h, P_H = 0.50)
M: XXXXXXXXXXXXXXXXXXXXXXXX   (24/24 h, P_M = 1.00, wraps at midnight)
Hr: 0    4    8    12   16   20   24
```

9.5 Game Value Verification

Expected coverage across the 50/50 mix:

$$\mathbb{E}[P_H] = \tfrac{1}{2}\times 0.75 + \tfrac{1}{2}\times 0.50 = 0.625 = P_H^*, \qquad \mathbb{E}[P_M] = \tfrac{1}{2}\times 0.75 + \tfrac{1}{2}\times 1.00 = 0.875 = P_M^*. \checkmark$$

Expected defender utility against each attack:

$$\mathbb{E}[U_d \mid \text{attack } H] = \tfrac{1}{2}\times 3.0 + \tfrac{1}{2}\times 0.0 = 1.5 = z^*, \qquad \mathbb{E}[U_d \mid \text{attack } M] = \tfrac{1}{2}\times 1.0 + \tfrac{1}{2}\times 2.0 = 1.5 = z^*.$$

The attacker is indifferent across the mix. The defender achieves game value $z^* = 3/2$, which is strictly better than playing either pure strategy alone:

Strategy A alone: $\min(3.0, 1.0) = 1.0 < z^*$, Strategy B alone: $\min(0.0, 2.0) = 0.0 < z^*$.

9.6 Minimum Fleet Size

Using (16) with $N_b = 2$, $T = 24$, $\mu = 3$, $F = 6$:

$$N_{\min}(\tfrac{1}{2}) = \left\lceil \frac{\frac{1}{2} \times 24 \times 2}{3 \times 6} \right\rceil = \left\lceil \frac{24}{18} \right\rceil = \lceil 1.33 \rceil = 2.$$

$$N_{\min}(1) = \left\lceil \frac{24 \times 2}{3 \times 6} \right\rceil = \left\lceil \frac{48}{18} \right\rceil = \lceil 2.67 \rceil = 3.$$

The current 2-aircraft fleet exactly meets the zero-loss threshold. Full guaranteed coverage requires 3 aircraft.

Algorithm 1 Stackelberg Patrol Planning

Require: Parameters $N, K, \{n_k, v_k\}, T, \tau_p, W, \mu, F$
Ensure: Mixed strategy s^*, game value z^*, operational timetables

$B_{\text{eff}} \leftarrow \lfloor NF/\tau_p \rfloor \cdot \tau_p$
// **Stage 1**
Compute $d_k \leftarrow n_k T/(2\mu v_k)$ for each k
$z^* \leftarrow (B_{\text{eff}} - TN_b/(2\mu))/\sum_k d_k$
$h_k^* \leftarrow d_k(z^* + v_k)$ for each k [or solve LP (4)]
Set $h_k^* \leftarrow 0$ for any type where $h_k^* < 0$; re-solve over remaining types
// **Stage 2**
Enumerate candidate strategies near $m_k^* = h_k^*/\tau_p$
Solve Phase-1 LP (9) $\rightarrow$ mixing probabilities s^*
Assert $\mathbf{1}^\top \varepsilon^* = 0$ (zero gap)
// **Stage 3**
for all pure strategy j with $s_j^* > 0$ **do**
    Distribute sorties across individual barriers (floor/ceil)
    Compute start times via (15)
    Assign sorties to aircraft (greedy, respecting gap G and budget F)
    Output timetable
end for
// **Goal 2**
Report $N_{\min}(1/2)$ and $N_{\min}(1)$ via (16)

10 Solution Algorithm

11 Why the Floor/Ceil Mixed Strategy is Worst-Case Optimal

Stage 2 produces a mixed strategy whose support contains pure strategies whose sortie allocations differ from each other by at most one sortie per barrier type. One might ask whether a more *diversified* mixing—one whose pure strategies vary by many sorties across barrier types—provides greater unpredictability to the adversary while preserving the same expected payoff. We show that any such diversification is suboptimal in the worst case: it necessarily exposes the defender to a lower realized utility in at least one period, and the floor/ceil mixing is the unique maximizer of the worst-case realized utility.

11.1 Definitions

Fix the optimal continuous allocation $h^* = (h_1^*, \ldots, h_K^*)$ from Stage 1. Define the *fractional sortie target* for type k as $m_k^* = h_k^*/\tau_p$. In the relevant case, m_k^* is not an integer for at least one k.

Definition 5 (Feasible mixed strategy)**.** *A* feasible mixed strategy *is a probability vector $s = (s_1, \ldots, s_J)$ with $s_j \geq 0$, $\sum_j s_j = 1$, together with a finite set of pure strategies $\{(m_{j1}, \ldots, m_{jK})\}_{j=1}^J$, where each m_{jk} is a non-negative integer and $\sum_k m_{jk} = N_{srt}$, satis-*

fying the coverage constraint:

$$\sum_{j=1}^{J} s_j \, m_{jk} \;=\; m_k^* \qquad \textit{for all } k = 1, \ldots, K.$$

Denote the set of all feasible mixed strategies by $\mathcal{S}^*$.

Every element of $\mathcal{S}^*$ achieves the same expected defender utility z^* by construction. What differs between elements of $\mathcal{S}^*$ is the *realized* utility in any single period.

Definition 6 (Realized utility and worst-case criterion)**.** *When pure strategy j is executed and the attacker strikes barrier type k, the realized defender utility is:*

$$U_d^{(j)}(k) \;=\; v_k \left(\frac{2\mu \, m_{jk} \, \tau_p}{n_k \, T} - 1 \right).$$

Since $U_d^{(j)}(k)$ is strictly increasing in m_{jk}, a period in which fewer sorties are assigned to type k is worse for the defender when the attacker strikes k. The worst-case realized utility *of a mixed strategy $s \in \mathcal{S}^*$ is:*

$$W(s) \;=\; \min_{\substack{j:\, s_j > 0 \\ k \in \{1,\ldots,K\}}} U_d^{(j)}(k).$$

11.2 The Key Lemma

Lemma 4 (Floor constraint on the support)**.** *For every feasible mixed strategy $s \in \mathcal{S}^*$ and every barrier type k, at least one pure strategy j in the support of s satisfies $m_{jk} \le \lfloor m_k^* \rfloor$.*

Proof. Fix any k. Suppose for contradiction that $m_{jk} \ge \lceil m_k^* \rceil$ for every j with $s_j > 0$. Since m_k^* is not an integer, $\lceil m_k^* \rceil > m_k^*$, and therefore:

$$\sum_{j:\, s_j > 0} s_j \, m_{jk} \;\ge\; \lceil m_k^* \rceil \cdot \sum_{j:\, s_j > 0} s_j \;=\; \lceil m_k^* \rceil \;>\; m_k^*.$$

This contradicts the coverage constraint $\sum_j s_j \, m_{jk} = m_k^*$. □ □

Lemma 4 says there is no escape from low-coverage periods: any valid mixing must include at least one pure strategy with sortie count at most $\lfloor m_k^* \rfloor$ on type k, for every type k.

11.3 Main Theorem

Theorem 5 (Floor/ceil optimality)**.** *Define the* floor bound *as:*

$$W^* \;=\; \min_{k=1,\ldots,K} \; v_k \left(\frac{2\mu \, \lfloor m_k^* \rfloor \, \tau_p}{n_k \, T} - 1 \right).$$

Then $W(s) \le W^*$ for all $s \in \mathcal{S}^*$, and the floor/ceil mixed strategy achieves $W(s_{FC}) = W^*$. Consequently,

$$\max_{s \in \mathcal{S}^*} W(s) \;=\; W^* \;=\; W(s_{FC}).$$

Proof. **Upper bound.** Fix any $s \in \mathcal{S}^*$. By Lemma 4, for each k there exists some $j(k)$ in the support of s with $m_{j(k),k} \le \lfloor m_k^* \rfloor$. Since $U_d^{(j)}(k)$ is strictly increasing in m_{jk}:

$$W(s) \;\le\; U_d^{(j(k))}(k) \;\le\; v_k\left(\frac{2\mu\,\lfloor m_k^* \rfloor\,\tau_p}{n_k T} - 1\right) \quad \text{for every } k.$$

Taking the minimum over k gives $W(s) \le W^*$.

Achievability. In the floor/ceil strategy s_{FC}, every pure strategy j in the support satisfies $m_{jk} \in \{\lfloor m_k^* \rfloor, \lceil m_k^* \rceil\}$ for all k. In particular, $m_{jk} \ge \lfloor m_k^* \rfloor$ for all j and k, so:

$$U_d^{(j)}(k) \;\ge\; v_k\left(\frac{2\mu\,\lfloor m_k^* \rfloor\,\tau_p}{n_k T} - 1\right) \quad \text{for all } j \text{ in the support and all } k.$$

Hence $W(s_{\mathrm{FC}}) \ge W^*$. Combined with the upper bound, $W(s_{\mathrm{FC}}) = W^*$. □ □

11.4 Why Diversification Cannot Help

Suppose a practitioner insists on a more varied mixing in which some pure strategy j' assigns $m_{j',k} < \lfloor m_k^* \rfloor$ to type k, hoping to increase unpredictability. To satisfy the coverage constraint, some other strategy j'' in the support must compensate with $m_{j'',k} > \lceil m_k^* \rceil$. While j'' gives higher coverage in the periods it is active, the adversary can simply ignore those periods and concentrate attacks in periods when j' is active. The realized utility in those periods is strictly below W^*, so $W(s) < W^*$.

Remark 11 (Adversary patience)**.** *The argument requires the adversary to be able to wait for a low-coverage period before attacking. If the adversary must commit to attacking a fixed barrier before observing which plan is active—a "committed attacker" model—then all elements of $\mathcal{S}^*$ are equivalent and diversification neither helps nor hurts. The Theorem applies to the* adaptive adversary *who observes the plan before attacking, or equivalently, who repeatedly attacks over many periods and exploits the lowest-coverage periods.*

Remark 12 (Integrality gap)**.** *The gap $W^* - z^*$ is the cost of discretizing sorties. For large N_{srt} (many sorties), m_k^* is nearly integer, the gap $\lfloor m_k^* \rfloor / m_k^* \to 1$, and $W^* \to z^*$. The floor/ceil construction makes this gap as small as it can possibly be.*

12 Comparison with Baseline Patrol Policies

We evaluate four standard patrol policies using the same defender utility framework and compare their game values against the Stackelberg optimal. Throughout, $B_{\mathrm{eff}} = 1{,}209\,\mathrm{h}$, $N_{\mathrm{srt}} = 403$ sorties, $\mu = 3$, $T = 720\,\mathrm{h}$, and barrier parameters are as in Table 1.

12.1 Baseline 1: Fixed Deterministic Patrol

Policy. Sorties are allocated equally across all ten barriers (40 sorties each, with three barriers receiving 41 to absorb the remainder), and start times follow a fixed published schedule.

Adversary exploit. An adversary who observes the fleet for two or three days learns the exact $\Delta_i \approx 18\,\mathrm{h}$ patrol clock for each barrier. The adversary then attacks in the $(W = 9)\,\mathrm{h}$ unprotected window immediately following a patrol. At the moment of attack the effective coverage probability is zero for the targeted barrier.

Defender utility.

$$P_k = 0 \quad \Longrightarrow \quad U_d(k) = v_k(2 \cdot 0 - 1) = -v_k \quad \forall\, k.$$

Game value $= \min_k U_d(k) = -v_H = -30$.

12.2 Baseline 2: Uniform Random Patrol

Policy. Sorties are allocated equally across all ten barriers (40–41 each), and start times are independently randomised each day using the phase draw of Section 7.2. No value-weighting is applied.

Allocation. Distributing 403 sorties uniformly: three barriers receive 41 sorties and seven receive 40. Assigning one extra sortie to each type gives $m_H = 161$, $m_M = 121$, $m_L = 121$ sorties, hence:

$$h_H = 483\,\text{h}, \quad h_M = 363\,\text{h}, \quad h_L = 363\,\text{h}.$$

Coverage and utility.

$$P_H = \frac{3 \times 483}{4 \times 720} = \frac{1{,}449}{2{,}880} \approx 0.5031, \qquad U_d(H) = 30(2 \times 0.5031 - 1) \approx +0.19,$$

$$P_M = \frac{3 \times 363}{3 \times 720} = \frac{1{,}089}{2{,}160} \approx 0.5042, \qquad U_d(M) = 20(2 \times 0.5042 - 1) \approx +0.17,$$

$$P_L = \frac{3 \times 363}{3 \times 720} \approx 0.5042, \qquad U_d(L) = 10(2 \times 0.5042 - 1) \approx +0.08.$$

Game value $= \min_k U_d(k) = U_d(L) \approx +0.08$. The adversary targets Low barriers. Despite positive utilities for all types, the policy is suboptimal because it ignores the value asymmetry: Low barriers yield less utility per unit of coverage, so equalising sorties leaves them as the weakest link.

12.3 Baseline 3: Value-Proportional Patrol

Policy. Sorties are allocated in proportion to the aggregate barrier value of each type, $n_k v_k$. Randomised start times are used.

Allocation.

$$\text{Total weight} = \sum_k n_k v_k = 4 \times 30 + 3 \times 20 + 3 \times 10 = 210.$$

$$m_H = \left\lfloor 403 \times \frac{120}{210} \right\rfloor = 230, \quad m_M = \left\lfloor 403 \times \frac{60}{210} \right\rfloor = 115, \quad m_L = 403 - 230 - 115 = 58.$$

Hence $h_H = 690\,\text{h}$, $h_M = 345\,\text{h}$, $h_L = 174\,\text{h}$.

Coverage and utility.

$$P_H = \frac{3 \times 690}{4 \times 720} \approx 0.719, \qquad U_d(H) = 30(2 \times 0.719 - 1) \approx +13.13,$$

$$P_M = \frac{3 \times 345}{3 \times 720} \approx 0.479, \qquad U_d(M) = 20(2 \times 0.479 - 1) \approx -0.83,$$

$$P_L = \frac{3 \times 174}{3 \times 720} \approx 0.242, \qquad U_d(L) = 10(2 \times 0.242 - 1) \approx -5.17.$$

Game value ≈ -5.17. H barriers are heavily over-protected while L barriers fall below the 50% breakeven coverage. The adversary exploits L.

12.4 Baseline 4: Greedy (Highest-Value First)

Policy. The budget is consumed greedily: fill High barriers to full coverage first, then Medium with the remainder, leaving Low unpatrolled. Randomised start times are used.

Allocation. Full coverage of $n_H = 4$ High barriers requires $h_H = n_H T/\mu = 4 \times 720/3 = 960$ h. Remaining budget: $1{,}209 - 960 = 249$ h allocated to M; $h_L = 0$.

Coverage and utility.

$$P_H = \frac{3 \times 960}{4 \times 720} = 1.0, \qquad U_d(H) = 30(2 \times 1 - 1) = +30,$$
$$P_M = \frac{3 \times 249}{3 \times 720} \approx 0.346, \qquad U_d(M) = 20(2 \times 0.346 - 1) \approx -6.17,$$
$$P_L = 0, \qquad U_d(L) = 10(2 \times 0 - 1) = -10.$$

Game value $= \min(-6.17, -10) = -10$. The adversary ignores the perfectly guarded H barriers and attacks L, which is completely unpatrolled.

12.5 Summary Table and Discussion

Table 5: Defender expected utility under five patrol policies. Game value $= \min_k U_d(k)$ when the adversary plays optimally.

Policy	h_H	h_M	h_L	$U_d(H)$	$U_d(M)$	$U_d(L)$	**Game value**
Fixed Deterministic	480	360	360	−30.00	−20.00	−10.00	−30.00
Uniform Random	483	363	363	+0.19	+0.17	+0.08	+0.08
Value-Proportional	690	345	174	+13.13	−0.83	−5.17	−5.17
Greedy (High-Value First)	960	249	0	+30.00	−6.17	−10.00	−10.00
Stackelberg Optimal	**482**	**362**	**365**	+0.13	+0.13	+0.13	**+0.13**

h_k = total flying hours on all barriers of type k; $U_d(k) = v_k(2P_k - 1)$ where $P_k = 3h_k/(n_k T)$.

Two observations stand out. First, only the Stackelberg solution achieves equal U_d across all types, making the adversary exactly indifferent. Every other randomised policy leaves at least one type below the others, providing a clear target. Second, the intuitive approaches (value-proportional and greedy) perform significantly worse than uniform random, because they create severely under-protected barriers in pursuit of protecting high-value ones. The Stackelberg framework avoids both failure modes by design: it explicitly optimises the worst-case utility, so no barrier type can be a preferred target.

Notation Summary

Symbol	Meaning
N, K	Number of aircraft; number of barrier types
n_k, v_k	Count and value of barriers of type k
$N_b = \sum_k n_k$	Total number of individual barriers
$T, \tau_p, \tau_s, W, \mu$	Planning horizon, patrol, safe, window, multiplier
$\tau_r, G = \tau_p + \tau_r$	Turnaround; minimum gap between sorties
$F, B, N_{\text{srt}}, B_{\text{eff}}$	Budget per aircraft; fleet total; integer sorties; effective budget
h_k	Total flying hours on barriers of type k
P_k	Coverage probability for one barrier of type k
$U_d(k;\, h)$	Defender expected utility when attacker strikes type k
z^*	Stackelberg game value (optimal worst-case defender utility)
$d_k = n_k T/(2\mu v_k)$	Sensitivity coefficient for type k
m_k, m_{jk}	Sortie count for type k (in strategy j)
A_{jk}	Flying hours on type k in pure strategy j
s_j	Mixing probability of pure strategy j
ε_k	Feasibility slack (coverage deficit) for type k in Stage 2
λ_k	Dual variable of Stage-2 coverage constraint (marginal coverage value)

References

[1] M. Tambe (Ed.), *Security and Game Theory: Algorithms, Deployed Systems, Lessons Learned*. Cambridge University Press, 2011.

[2] P. Paruchuri, J. P. Pearce, J. Marecki, M. Tambe, F. Ordonez, and S. Kraus, "Playing games for security: An efficient exact algorithm for solving Bayesian Stackelberg games," *Proc. AAMAS*, pp. 895–902, 2008.

[3] D. Korzhyk, V. Conitzer, and R. Parr, "Solving Stackelberg games with uncertain observability," *Proc. AAMAS*, 2011.